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Accenture Song

# Community Evacuation Resource Matcher (CERM)

Route 1: Accelerating Equitable Evacuations

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Georgia Institute of Technology


**Ramblin' Pathfinders**

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*Anthony Mitchell Sr. stayed behind because his son Justin, who had cerebral palsy, could not leave without him. They called for help. They waited. The Eaton Fire did not wait with them.*

People of vulnerable communities are most at risk during wildfires. Preventable deaths result from lack of coordination, and neighbors and organizations with resources have no way to find the people who need them.

**CERM connects people who need evacuation help with community members who can provide it in real time, before the fire arrives.**

# The Problem

## The Scale

| | |
|---|---|
| **3.7M** | 3.7M Californians live in High and Very High Fire Hazard Severity Zones and many have no reliable way to call for community help when evacuation orders drop (Cal Fire / Direct Relief) |
| **2.5×** | Elderly people are 2.5 times more likely to die in a fire compared to the general population (FEMA). People with a disability are 2.5 times more likely to be displaced by a natural disaster (NBC News). |
| **1 in 6** | About 1 in 6 Americans over 50 cannot evacuate with outside help. Half need assistance from someone outside their household (AARP). |

## The Cost of Inaction

- Globally, only 31% of people with disabilities always have someone to help them evacuate, and 13% never have anyone (UN DESA)
- Many elderly and disabled residents have no nearby family, can't afford to leave, and are dependent on a coordination infrastructure that doesn't exist.
- Key reason a disproportionate number of disabled persons suffer and die in disasters: their needs are ignored by the official planning process.
- The community layer has no platform to fill that gap. Without CERM, that remains true in the next fire.



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# Why Now

## *Four Converging Forces Have Created This Window*

| | |
|---|---|
| 1 | Real-Time Wildfire Data Is Now Public & Reliable<br>• WatchDuty served 8 million users in 1 week during January 2025 LA fires and maintained 100% uptime<br>• Displayed as the primary live feed in the LA Emergency Operations Center<br>• Fire perimeter, evacuation zones, and incident data now available as a structured source for tools like CERM |
| 2 | Census Tract Data Enables Hyper-Local Vulnerability Mapping<br>• US Census data on vehicle ownership, disability, age, and poverty is publicly available at the tract level<br>• Can be overlaid with active wildfire data in real time<br>• Allows CERM to surface highest-risk neighborhoods and precisely direct community support |
| 3 | Climate Urgency Has Made Wildfire a Year-Round Policy Priority<br>• CA wildfires are no longer seasonal. Climate change has extended fire conditions year-round<br>• FEMA allocated $1+ billion in technology grants for state & local emergency management in FY2026, a direct funding pathway for CERM |
| 4 | Community Willingness to Help Exists But the Platform Does Not<br>• Faith organizations, mutual aid groups, ride-share volunteers, and community nonprofits are willing to support evacuations<br>• No structured way currently exists to match them with help in real time<br>• CERM closes that gap with a purpose-built community coordination platform |



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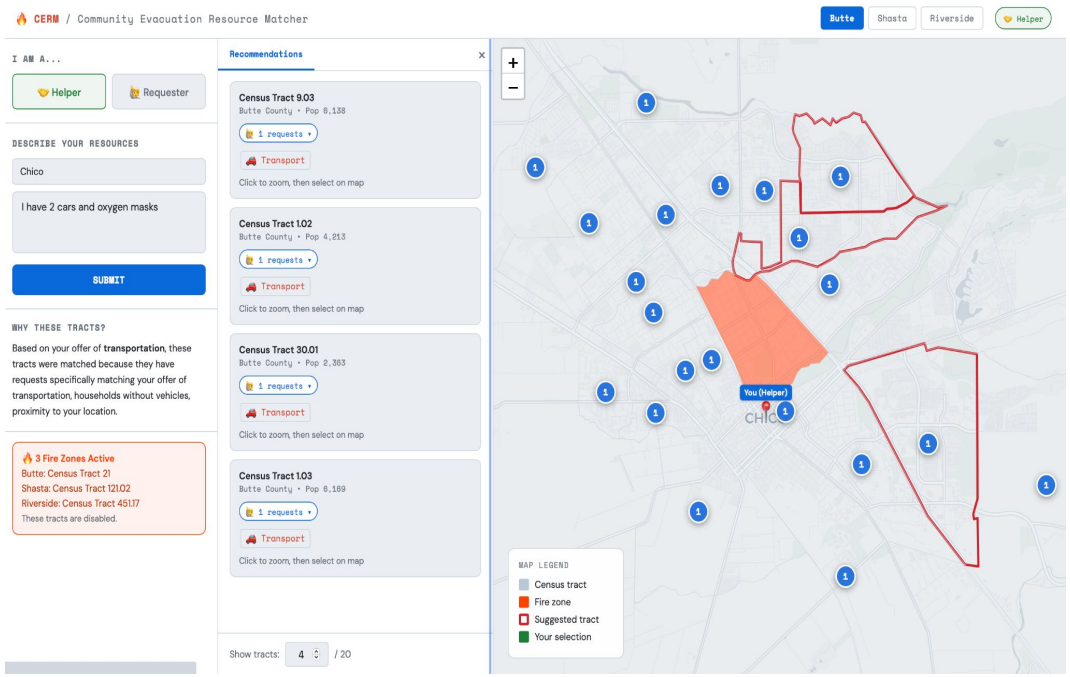
# The Solution

## Community Evacuation Resource Matcher

CERM matches community helpers to the highest-need areas during a wildfire, using demographic vulnerability data to surface where help is needed most, and live requests to match helpers' specific resources to the people who need them.




**If you need help:**

- Open the app and submit a request (a ride, pet assistance, a wheelchair-accessible vehicle, or any other support)
- Your request is instantly visible to community organizations in your area

**If you're an organization or volunteer:**

- View a map showing which areas have the highest concentration of vulnerable residents and are near to you
- See a live dashboard of open requests, automatically sorted by type
- Dispatch resources directly (no phone trees, no guesswork)

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# How It Works

## THREE-STEP LOGIC — FROM RAW DATA TO LIFE-SAVING DECISIONS

**STEP 1**

**We Take This Data**

CERM ingests census tract vulnerability data (elderly population, disability rates, households without vehicles), active fire zone overlays modeled on wildfire incident data, and live requests submitted by people in evacuation zones needing community support.

▼ We Identify Where Help is Needed Most ▼

**STEP 2**

**We Identify Where Help Is Needed Most**

CERM identifies which neighborhoods need help most based on who lives there (elderly residents, people with disabilities, and households without vehicles). Within those areas, individual requests are automatically sorted by type and matched to the nearest available helper.

▼ We Produce This Output ▼

**STEP 3**

**We Produce An Actionable Output**

A helper opens CERM and sees a high-need neighborhood nearby (many residents have no car, most are elderly, and 14 people need wheelchair transport). They respond in minutes. No phone calls. No guesswork.



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# Traction

## EVIDENCE THE PROBLEM IS REAL AND THE SOLUTION IS NEEDED

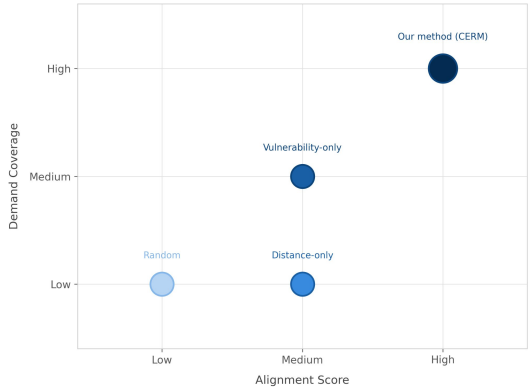
*Expert Validation: Multiple Community Preparedness Officials Interviewed*

The Ramblin' Pathfinders team conducted structured interviews with community preparedness officials across California prior to and during the development of this prototype. Their feedback was consistent across all these conversations:

| | |
|---|---|
| **Finding 1** | No reliable coordination mechanism exists for community help during wildfire evacuations. Officials described current processes as fragmented: phone trees, social media, and personal networks that collapse under pressure. |
| **Finding 2** | People with disabilities, the elderly, and individuals without access to private transportation represent the highest-risk evacuation population and are the least served by existing systems. |
| **Finding 3** | A tool that directs community helpers to the highest-need geographic areas informed by who lives there (not just who asks) and connects them to individual requests that would meet a clearly unmet need in emergency infrastructure. |

### MARKET SIGNAL (EXTERNAL VALIDATION)

In April 2025, the LA County Board of Supervisors unanimously approved a motion to study a vulnerable population registry, so that first responders know where high-need residents live before a disaster strikes. CERM already builds that capability at the community level. The government is studying what we have already built.



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# Market Opportunity

## Who Uses This

- **Requesters:** Primarily elderly residents, people with disabilities, and households without vehicles in wildfire evacuation zones
- **Helpers:** Nonprofits, faith organizations, mutual aid networks, neighbors, and community volunteers who already show up during evacuations but have no structured way to find where they are needed most.

## Who Pays This

| | |
|---|---|
| **Government** | County emergency management offices and FEMA grant programs, including the $1B BRIC program (FY2024–25), a direct funding pathway for community-layer disaster tools. |
| **Nonprofits** | Free at early stage, funded through government grant partnerships until adoption data supports a sustainable model. |

## The Market

The emergency response software market is growing from $5.35 billion today to $13.76 billion by 2034. No existing tool fills the gap CERM addresses. Every app tells you where the fire is. CERM tells you who needs help most and gets volunteers there.

## HOW IT REACHES THE PEOPLE WHO NEED IT

**Phase 1:** Pilot in Butte, Shasta, and Riverside: the exact counties our vulnerability analysis already covers.

**Phase 2:** California-wide via CalOES integration and FEMA grant funding.

**Phase 3:** National community coordination layer for wildfire-prone communities across the US.



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# The Team

## RAMBLIN' PATHFINDERS – WiDS APPRENTICESHIP POD

The Ramblin' Pathfinders team brings together a rare combination of data science depth and personal experience in wildfire evacuation situations. This is the exact pairing that separates prototype-stage tools that get adopted from those that stay on a slide deck.

Simran Mallik

Data scientist with 2 years of industry experience and an MS at Georgia Tech. A Bay Area resident during the 2020 fires, Simran experienced firsthand the absence of community coordination during a wildfire emergency, and helped build CERM to fill it.

Riya Bharathwaj

As the team's data engineer, Riya brings hands-on experience building map interfaces and LLM data pipelines, with a background in healthcare ML research that directly informs how CERM models and prioritizes community vulnerability.

Anand Acharya

Senior advisor on the team with 20+ years experience in building and scaling Data & AI products. Will be providing counsel on commercial model, business partners and bringing CERM to market.

Ting Ya Chang

Former AI solutions engineer and COO of an education startup, now an MS Analytics student at Georgia Tech. Ting specializes in building user-facing applications and scalable systems, supporting CERM's usability and real-world implementation.

Saehee Eom

AI-focused graduate student at Georgia Tech with experience building data-driven systems. Working on CERM sparked Saehee's interest in applying AI to disaster resilience, with the goal of building tools that can directly support communities during emergencies.

Tanmayee Koli

MS Analytics student at Georgia Tech with experience in healthcare research, Agentic AI, and manufacturing. Tanmayee's experience in patient- and client-focused work helps shape CERM into a solutions that is data-driven and grounded in real-world needs.



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# The Ask

## We Are Asking for $529,000 in Investment

*We're seeking a $529,000 investment to take CERM from prototype to production: funding engineering, pilot partnerships, and data infrastructure to deploy in at least one California county before the 2026 fire season. This isn't a concept pitch, it's a working prototype addressing a validated problem, backed by a team that has already shown up to solve it.*

## Funding Uses for Investment

| Category | Subcategory | Description | Cost ($) |
|---|---|---|---|
| **Engineering** | Web App & Matching Engineer | Frontend, matching pipeline, cloud hosting and failover | $150,000 |
| | Data & Fire Integration Engineer | Census/vulnerability data pipeline and live fire data integration | $150,000 |
| **Government Partnerships** | County & Agency Outreach Lead | Secure county agreements in Butte, Shasta, and Riverside | $60,000 |
| **Marketing** | Community & County Outreach | Multilingual outreach across 3 counties; scale to California | $35,000 |
| | Digital and Paid Channels | Social ads, SEO press outreach, community events, materials | $20,000 |
| **Security & Accessibility** | Auth, Location & Multilingual | Authentication, location access, multilingual support for all users | $30,000 |
| **Legal & Compliance** | Privacy, Terms & CalOES | Privacy policy, terms of service, CalOES coordination | $30,000 |
| **Platform Costs** | Cloud & Infrastructure | AWS/GCP hosting, database, CDN, failover, monitoring (12mo.) | $24,000 |
| **Finance & Accounting** | Bookkeeping & Reporting | CPA, Grant reporting, audit prep, payroll admin (12mo.) | $18,000 |
| **Travel** | County & Partner Site Visit | Team Travel to Butte, Shasta, Riverside; stakeholder meetings | $12,000 |



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